SegDINO3D: 3D Instance Segmentation with 2D Features

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Problem & Approach

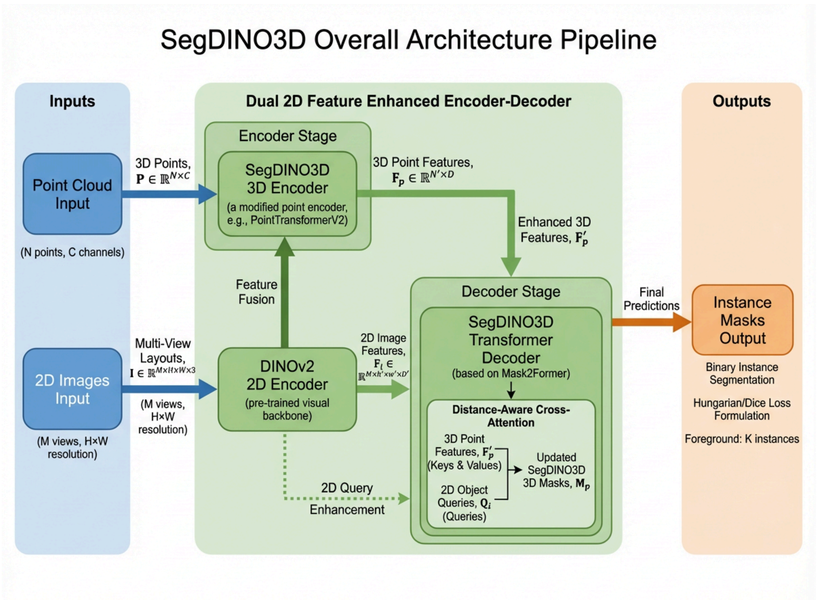
SegDINO3D addresses limited 3D training data by leveraging **pre-trained 2D detection models** to enrich 3D point clouds with both image-level and object-level features.

- **Transformer encoder-decoder** framework for 3D instance segmentation
- **Cross-attention** from 3D queries to 2D object queries

Box-Modulated Cross-Attention

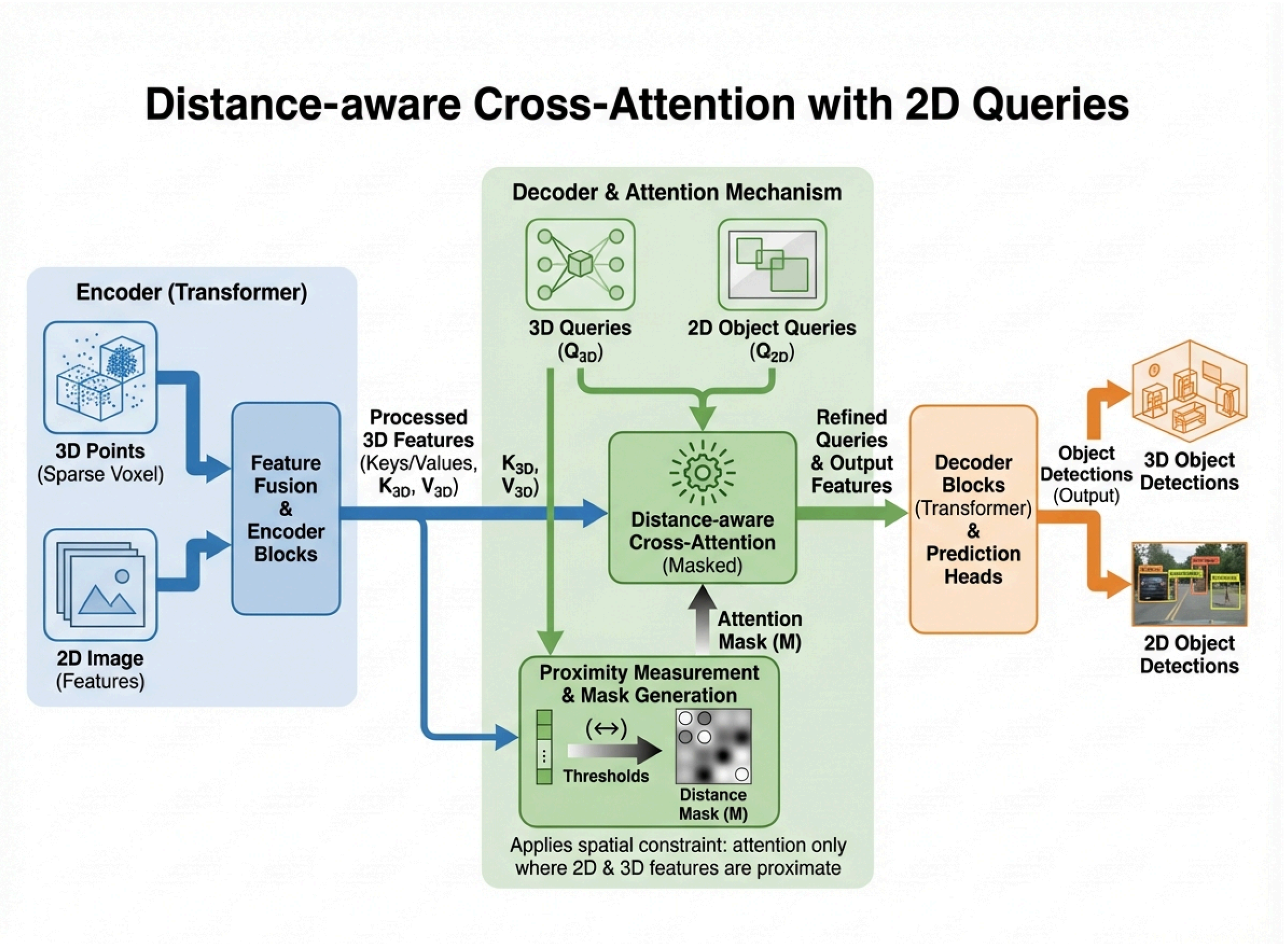
Box-modulated cross-attention leverages explicit 3D bounding box dimensions to overcome spatial confusion in conventional mask...

- Adds **positional component** Q_P encoding (x,y,z,l,w,h) coordinates
- **Sinusoidal encoding** with temperature T=20 for positional similarity
- **Modulation function** adapts attention maps based on object size



Performance Comparison

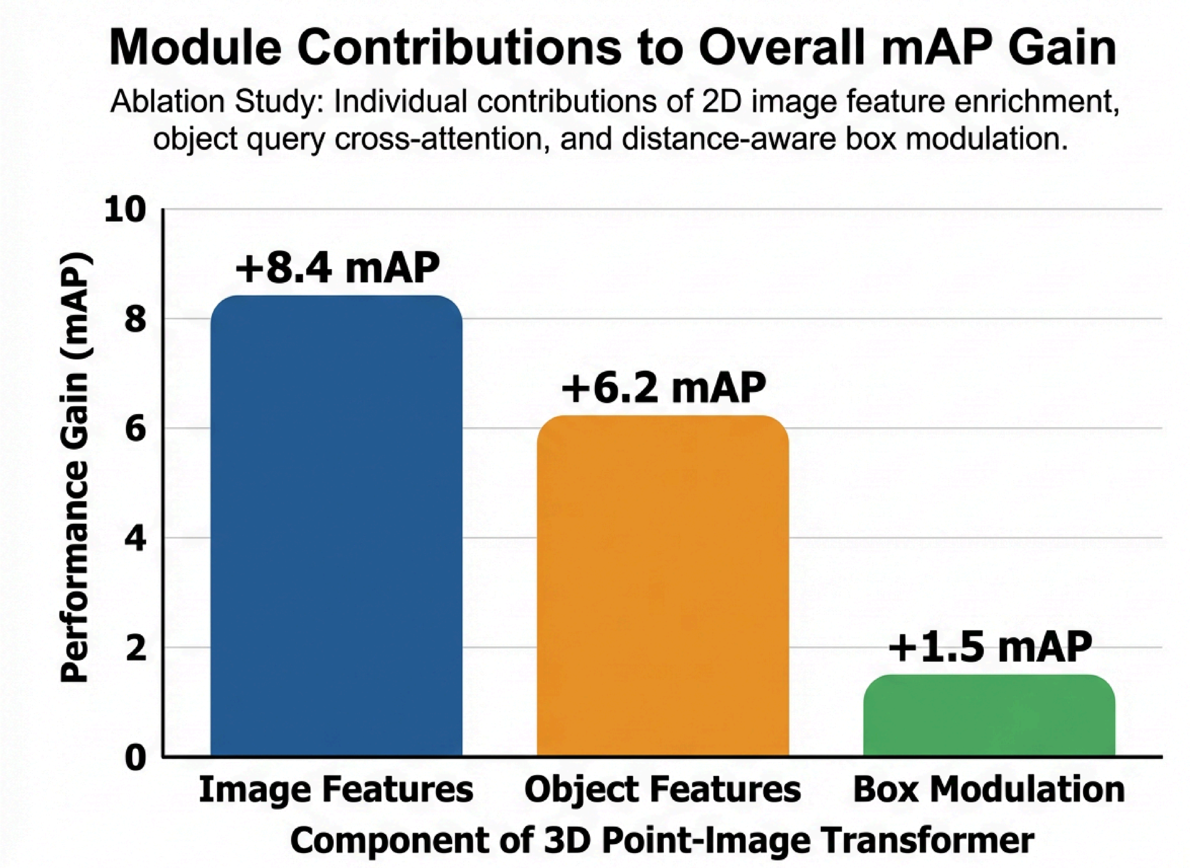
- **+8.7 mAP** improvement over prior best on ScanNet200
- **+4.7 mAP** boost over baseline on ScanNetv2
- Tail-split performance **36.2 mAP** exceeds common-split baselines



The Challenge

3D instance segmentation remains challenging despite advances in reconstruction, as most methods ignore rich **2D semantic features** from accompanying images used to build point clouds.

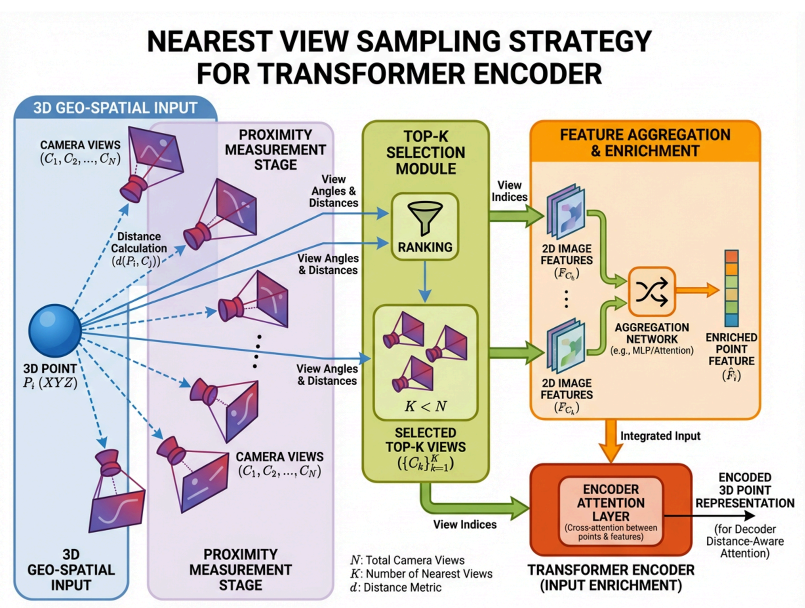
- **2D foundation models** trained on massive datasets provide richer semantics
- Existing methods degrade **2D features** through limited 3D encoders



Distance-Aware Cross-Attention

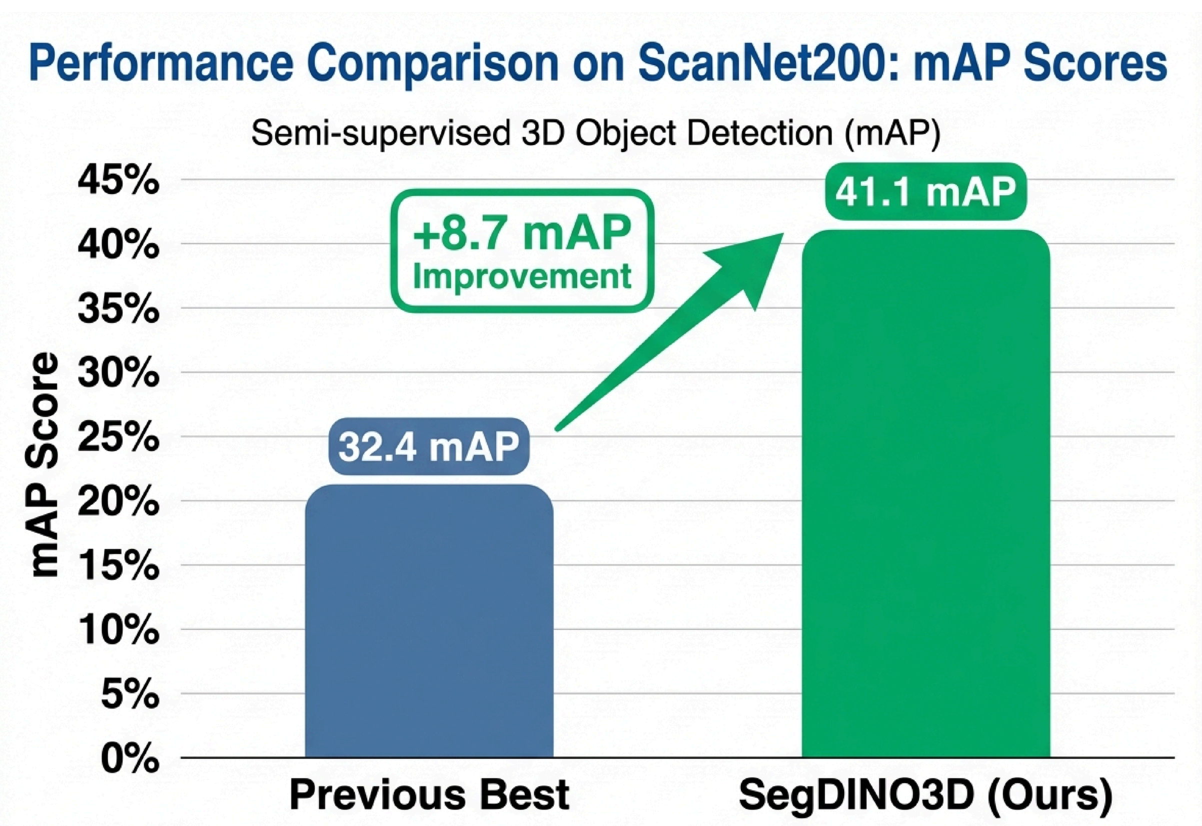
SegDINO3D uses **distance-aware cross-attention** to selectively fuse 3D object queries with nearby 2D object features.

- **Spatial proximity mask** prevents confusion from distant 2D objects
- **L2 distance** between superpoints and 2D objects determines attention
- **Cross-attention** optimizes 3D queries



Ablation Analysis

- **Image-level features** boost performance by **+8.4 mAP**
- **Object-level features** add **+6.2 mAP** improvement
- **Combined features** yield additional **+3.1 mAP** gain



SegDINO3D, a Transformer encoder-decoder, achieves a significant gain by enriching 3D data with 2D features and using distance-aware cross-attention with object queries.

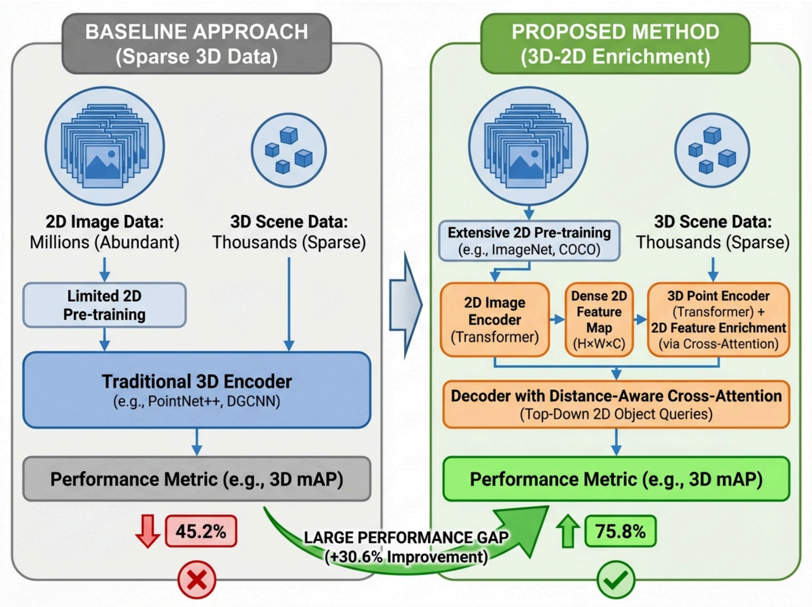
Feature Preparation Pipeline

SegDINO3D constructs enriched 3D features and compact 2D object queries through **Nearest View Sampling** and strategic object selection.

- **Top-3** nearest views selected per 3D point

****2D Object Query Construction****

- **PAM algorithm** determines robust 3D object centers
- **2048 queries** retained per scene for efficiency



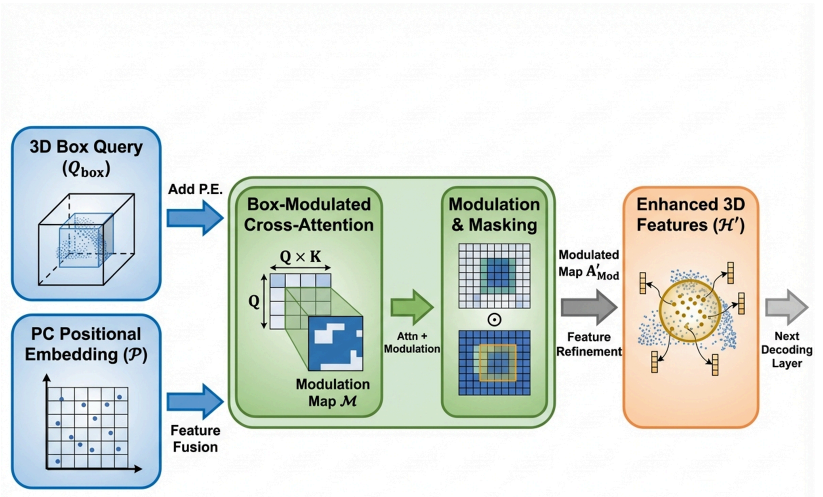
SegDINO3D Architecture

SegDINO3D fuses 2D DETR features with 3D point clouds through **Nearest View Sampling** and **Distance-Aware Cross-Attention**.

- Frozen **2D DETR** extracts features from **V posed RGB images**
- **Nearest View Sampling** maps 2D features to 3D points

****Decoder Stage****

- **Transformer decoder** with **Box-Modulated Cross-Attention** refines object queries



Experimental Setup

SegDINO3D is evaluated on **ScanNetv2** and **ScanNet200** datasets using standard **mAP** metrics.

- **ScanNet200**: 198 instance classes in head/common/tail splits
- **5× faster** convergence: 88 epochs vs 432
- Single **RTX 3090** GPU for training and inference